Detecting departure and destination aerodromes from ADS-B

Version 6.1 — the altitude cut, and which datum it is measured against

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2026-08-23

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1 What this version changes

Version 6 tuned `trend`'s altitude cut and left its *datum* alone. This version changes the datum, and measures what that is worth.

`trend` admitted a state vector to its vote when the aircraft was below a **flight level** — pressure altitude above the standard datum, the same number everywhere. `endpoint` had never worked that way: since version 6 its height gate has been measured **above the aerodrome's own field elevation**, on the stated grounds that "a fixed altitude cut-off means nothing at an aerodrome sitting at 5,000 ft". Step 04's ground-phase detection moved to the same datum for the same reason.

`trend` was the last of the three still referenced to sea level. That is the asymmetry this version removes.

The consequence is not inaccuracy but **silence**, and it falls unevenly. At FL60 an aerodrome at sea level offers 6,000 ft of climb and descent for the vote to read. Madrid, at 1,998 ft, offers about 4,100. Ankara, at 3,125 ft, about 2,975. The altitude trace has progressively less room to rise or fall *clearly*, so the method abstains more often — and it abstains in proportion to how high the aerodrome sits.

> **i** What this change cannot do
>
> `trend` votes on the **sign** of a smoothed altitude change. Adding a constant offset to every altitude in a track cannot flip a sign, so the field elevation changes only *which samples are admitted to vote*, never how a vote is cast.
> This is therefore a **coverage** result first. Any movement in accuracy is a second-order consequence of answering more flights, not of discriminating better between them. The report is written to keep those apart.

2 How a flight gets its aerodromes

Version 6 describes the three methods and the reasoning behind them; this section does not repeat that. What it adds is the part version 6 states only in prose — the **order** the pipeline does things in, and which datum each altitude gate is measured against — because the change this version makes is a change to that order.

Both methods answer two questions per track, separately: which aerodrome the flight departed from (`ADEP`) and which it arrived at (`ADES`). The shipped configuration takes departures from `endpoint` and arrivals from `trend`.

2.1 The steps

Every box in the diagrams below is one of these.

S1 — Read the track. One aircraft's state vectors for one flight, already split into a track by step 02 and cleaned by step 02a.

S2 — Smooth the altitude. A centred five-sample rolling mean over barometric altitude, computed over the *whole* track before any altitude cut. Smoothing after the cut would truncate the window exactly where altitude is changing fastest, which is the worst place to lose half of it.

S3 — Pre-filter on pressure altitude. A cheap, deliberately loose cut whose only job is to keep the aerodrome join affordable. Its ceiling is the configured cap *plus the highest field elevation in the reference*, so it cannot discard a sample that the exact test at S6 would have kept.

S4 — Join the aerodrome zones. Each sample is matched to candidate aerodromes through an H3 index at resolution 7, then narrowed to those within the detection radius by exact great-circle distance.

S5 — Attach the field elevation. Each candidate aerodrome's elevation, from OurAirports. A candidate whose elevation is unknown gets zero, which reduces the next step to the sea-level behaviour rather than dropping the aerodrome.

S6 — Cut on altitude. *This is the step this version changes.* Samples above the ceiling are excluded from the vote. `trend` now measures that ceiling **above field elevation**; before this version it was a flight level.

S7 — Vote. Each surviving sample votes on the sign of its smoothed altitude change: rising for a departure, falling for an arrival.

S8 — Decide, or abstain. If one direction leads the other by more than the margin, the track is departing or arriving. Otherwise `trend` says nothing.

E1 — Take the endpoint fix. `endpoint` reads one sample, not many: the track's first fix for a departure, its last for an arrival.

E2 — Test the gates. The fix must be within the radius, and either flagged on the ground or no higher than the height gate **above field elevation**.

E3 — Rank the survivors. Nearest wins, on exact great-circle distance, plus a penalty for aerodromes without scheduled service so that a military or general-aviation field must be *clearly* nearest rather than merely nearest.

E4 — Or say where it came from. Where the gates fail and the fix sits at the edge of the observed area, `endpoint` reports **out-of-area** rather than guessing. Precedence is aerodrome first, border second: Ponta Delgada lies about 8 NM inside the western edge, and the other order labelled its departures out-of-area with the aircraft still on the runway.

2.2 `trend`, arrivals

Step	What it does
S1	Read the track's state vectors
S2	Smooth `baro_altitude` over a centred 5-sample window, before any cut
S3	Pre-filter on pressure altitude at the cap plus the highest field elevation
S4	Join the H3 aerodrome zones at resolution 7, then cut on exact distance
S5	Attach the candidate aerodrome's field elevation; unknown becomes zero
S6	**Cut on height above field elevation** — the step this version changes
S7	Each surviving sample votes on the sign of its smoothed altitude change
S8	If descending votes lead by the margin the track is arriving; else *undetermined*
E3	Rank survivors on distance plus the scheduled-service penalty; nearest becomes `ADES`

2.3 `trend`, departures

The same machinery reading the same samples; only the direction of the winning vote differs. `trend` has **no out-of-area concept** — it names an aerodrome or says nothing — which is why the shipped configuration does not use it for departures.

Identical to the arrival table above, with one difference: at **S8** the track is a departure when *climbing* votes lead by the margin. Failing that, `trend` returns *undetermined* — it has no out-of-area answer available to it.

2.4 `endpoint`, departures

Step	What it does
S1	Read the track's state vectors
E1	Take the **first** fix — one sample, not many
S4	Find candidate aerodromes within the radius
S5	Attach each candidate's field elevation
E2	Gate: within the radius **and** either on the ground or **no higher than the gate above field elevation**
E3	Passing: rank on distance plus the scheduled-service penalty; nearest becomes `ADEP`
E4	Failing, with the fix at the edge of the observed area: **out-of-area**. Otherwise *undetermined*

Precedence is aerodrome first, border second — the other order labelled Ponta Delgada's departures out-of-area with the aircraft still on the runway.

2.5 `endpoint`, arrivals

Identical to the departure table above, with one difference: at **E1** the method reads the track's **last** fix rather than its first.

2.6 Where the two datums part company

Only **S6** and **E2** read an altitude, and until this version they read it against different references:

Test	Threshold	Datum, before v6.1	Datum, from v6.1
`trend` S6	`trend_max_height_ft`	flight level (sea level)	**above field elevation**
`endpoint` E2	`endpoint_height_ft`	above field elevation	above field elevation
step 04 ground phase	`phase_ground_ceiling_ft`	above field elevation	above field elevation
`nearest`	*none*	—	—

`nearest` applies no altitude condition at all, on any datum. That is deliberate: it is the naive `traffic`-faithful baseline the other methods are scored against, and giving it a gate would defeat its purpose.

! A flight level is not a height

`FL60` is not the same cut as `6,000 ft`. The pipeline derives the flight level with an integer cast, so `flight_level <= 60` admits everything below **6,100 ft**. The datum comparison in this report therefore holds the ceiling at 6,100 ft above field, not 6,000 — otherwise it would move the ceiling and the datum at the same time and could not attribute the result to either.

3 Results

3.1 The band breakdown, read first

This section comes before the headline deliberately. The headline is a single pair of numbers and it is consistent with several explanations; the band breakdown is the only part of this study that can distinguish between them. Read in the other order, it becomes a search for confirmation.

The prediction the design committed to, before any of this was run: **no movement at sea-level aerodromes, and gains concentrated at elevated ones.**

Table 1: Arrivals, 2025 sample. Δ is field minus sea level.

Band	Aerodromes	Truth	Correct MSL	Correct field	Δ	Δ less top mover	Δ less busiest	Δ relative
<500	490	66,612	49,514	49,571	57	33	57	+0.12%
500-1500	164	15,285	12,593	12,621	28	17	28	+0.22%
1500-3000	77	4,434	1,919	2,028	109	70	110	+5.68%
>3000	26	705	5	5	0	0	0	+0.00%
unknown	49	8,080	0	0	0	0	0	+0.00%

The effect is where it was predicted to be. Measured against each band's own base of correct arrivals — which is the honest denominator, since the bands differ in size by two orders of magnitude — the sea-level control band moves by +0.12% and the 1,500–3,000 ft band by **+5.7%**. That is a difference of roughly forty-fold between the band where the datum change is arithmetically a no-op and the band where it bites hardest.

It also survives both robustness checks. The census run before this study began showed each treatment band resting on a single aerodrome, so a band mean alone could not separate an elevation effect from a Madrid effect. Removing the largest mover leaves +70 of the +109; removing Madrid, the busiest field in the band, leaves +110 — very slightly *more*. The result is not one aerodrome's result.

i The control that had to hold, and did

Departures move by exactly zero in every band. The shipped configuration takes departures from `endpoint`, which never reads `trend`'s altitude cut — so a non-zero number anywhere in that row would have meant this change was touching something it has no business touching. It is the cheapest available check that the intervention is the one described.

3.2 What it costs

Table 2: Arrivals through `process_dai` itself, 2025 sample.

Run	Coverage	Accuracy	Correct	Wrong	Score (k=2)
datum_msl	68.29%	98.58%	64,031	923	62,185
datum_field	68.60%	98.42%	64,225	1,028	62,169
legacy	66.99%	98.19%	62,564	1,151	60,262

Moving to the field datum buys **194 more correct arrivals and 105 more wrong ones**. Coverage rises 0.31 pp; accuracy falls 0.16 pp; the score at $k = 2$ moves from 62,185 to 62,169 — sixteen points the wrong way. In net terms it is close to a wash, tilted toward answering more flights and being fractionally less certain about each.

3.3 Why it ships anyway

The adoption criteria fixed in the design were: coverage up by at least 0.5 pp, accuracy not falling, and the gain concentrated at elevated fields. **The third holds and the first two do not.** On a cost-benefit reading, that would be the end of it.

The change is adopted regardless, and the reason is worth stating plainly because it is not a benchmark result.

A ceiling measured from sea level is *not the same test at different aerodromes.* At Amsterdam it admits 6,000 ft of climb and descent; at Ankara, 2,975; at Erzurum, roughly 240. Nobody chose that gradient — it is an artefact of which datum the threshold happened to be written against, and it makes the method's willingness to answer depend on the terrain under the aerodrome rather than on the evidence in the trajectory. `endpoint` was corrected on exactly this reasoning in version 6, and step 04's ground detection in the same release. `trend` was the last one left.

So this is a **correctness fix, not an optimisation**, and it is not the benchmark's place to veto it. What the benchmark is for is telling us what the fix costs, which it has done: a third of a percentage point of coverage gained, a sixth of a point of accuracy given up, and the movement concentrated where the mechanism predicts. That is the honest accounting, and it is reported here rather than buried because a change adopted on principle still has to declare its price.

The measured null in criteria 1 and 2 does carry one design implication: the ceiling itself, inherited from a sweep conducted on the datum now abandoned, is very likely no longer optimal. Section 4 returns to that.

3.4 The band that should have gained most, and did not

The `>3000` band is the one where a fixed sea-level ceiling ought to hurt worst, and it shows **no change at all: 5 correct arrivals out of 705, on both datums.**

That is not a null result about the datum. It is `trend` failing at those aerodromes for a different reason — a recall of 0.7% is not a method being mis-tuned, it is a method receiving almost nothing to work with. The likeliest cause is ADS-B coverage over the Anatolian plateau, where most of this band's traffic sits; an aerodrome the feed barely sees cannot be named on any datum.

This matters for how the change should be argued elsewhere. The motivating example in the design was Erzurum at 5,763 ft, where a FL60 cut leaves about 240 ft in which to vote. That example is arithmetically correct and doubly unrepresentative: such fields are rare in the sample, **and** they sit in the one band where the method is independently broken. The population that actually benefits is the far less dramatic 1,500–3,000 ft — Madrid, Ankara, Sofia, Tbilisi — where a FL60 cut still leaves three to four thousand feet of usable climb and descent.

3.5 The ceiling survives the move

`trend_max_height_ft` ships at the FL60 equivalent because that is what the datum comparison had to hold fixed to isolate one variable. That left an obvious worry: FL60 was tuned by V6 *on the datum this version abandons*, so there was no reason its optimum should have survived. The two sweeps settle it.

Table 3: Arrivals, shipped geometry (30 NM, margin 0, penalty 10 NM). Both curves swept out of the same vote cache.

Datum	Ceiling	Coverage	Accuracy	Score
above field	2,000 ft	65.73%	97.92%	58,625
above field	3,000 ft	67.31%	97.78%	59,748
above field	4,000 ft	68.64%	97.33%	60,067
above field	6,100 ft	70.35%	96.98%	60,857
above field	8,000 ft	71.45%	96.11%	60,026
above field	10,000 ft	72.84%	94.53%	57,918
above field	12,000 ft	73.84%	93.21%	55,924
flight level	FL20	64.87%	98.12%	58,225
flight level	FL30	66.92%	97.95%	59,750
flight level	FL40	68.14%	97.78%	60,492
flight level	FL60	70.01%	97.18%	60,952
flight level	FL80	71.18%	96.36%	60,309
flight level	FL100	72.53%	94.89%	58,409
flight level	FL120	73.68%	93.47%	56,354
flight level	FL150	74.47%	91.95%	53,721
flight level	FL200	75.45%	89.54%	49,248

Each datum peaks at the equivalent ceiling — 6,100 ft above field, FL60 on sea level — and 6,100 ft wins at *every* radius from 20 NM outward, production's 30 NM included. The shipped value is therefore tuned rather than inherited, and no change is indicated.

Two honest limits. The grid is coarse near the peak (4,000 → 6,100 → 8,000), so the true optimum could sit anywhere in that span; and these are harness numbers, which V6 showed can differ from the pipeline's in level even when they agree in shape. What the sweep establishes is the *shape*: an interior optimum, falling away on both sides, at the value already shipping.

3.6 Why the aggregate was never going to show much

The two curves are also nearly identical in level — 60,952 against 60,857 at the shipped geometry, and 61,069 against 61,056 at each datum's own best cell. That looks like a change doing nothing, and it is worth being precise about why it is not.

Roughly seventy per cent of the ground-truth arrivals sit in the `<500 ft` band, where the two datums are the same test by construction. A change that cannot move the majority of the sample cannot move an average over it, however well it works on the rest. The aggregate score is diluted by design, and judging a targeted correctness fix by it is using the wrong instrument — the reading is dominated by the aerodromes the fix was never about.

That is precisely why Section 3.1 exists and why it is read first. The signal is in the 1,500–3,000 ft band, and the aggregate is where that signal goes to be averaged away.

4 What is still open

The second period confirms through the sweeps, not the pipeline. Everything above is 2025-06-05…07. The 2024 sample cannot run the pipeline arms at all: `process_dai` reads `h3_res_7` off the track table, and the 2024 tracks were built before this study indexed them, so the column is absent. V6 met the same wall and ran the pipeline on 2025 alone.

What 2024 *can* do is the paired sweeps, out of a vote cache built with the index computed at read time — so the second period tests whether the ceiling and the datum behave the same way there, which is the part most at risk of being a one-sample artefact. Those two jobs are the whole of this period's evidence, and the report should say so rather than implying a symmetry the data does not support.

The `>3000` band needs a different question. A recall of 0.7% on both datums is a coverage problem upstream of detection, and no amount of threshold work will move it. Whether that is receiver geometry over the Anatolian plateau or something in ingestion is worth knowing, but it is not this study's question.